# GENERALIZED CLAIMS PROCESSING SOP

## Enterprise Claims Processing Standard

A common procedure for healthcare, banking/fraud, ATM, payment, property, liability and other claim types

| Document Attribute | Value |
|---|---|
| Document Type | Enterprise Standard Operating Procedure |
| Process | End-to-end claim intake, assessment, investigation, decision, settlement and closure |
| Applicability | Insurance and financial-services claim operations; adaptable to other claim domains |
| Operating Model | Rules-based processing, contextual analysis and human review where required |
| Primary Objective | Resolve claims consistently, accurately, transparently and within applicable service commitments |
| Version | 1.1 |
| Status | Updated baseline - Location Intelligence integrated |

## Purpose

This SOP defines a generalized claim-processing procedure that can be applied across different claim domains. It establishes a common lifecycle while allowing domain-specific rules, evidence, calculations, approvals, communications and location-intelligence controls to be plugged into the relevant processing branch. The procedure is intentionally domain-neutral. A healthcare claim may involve member, provider, clinical, billing and geographic information, while a banking or fraud claim may involve customer, account, transaction, channel, device and location information. Both follow the same core lifecycle: intake, validation, classification, investigation or assessment, decision, financial processing, communication and closure.

## Scope

- New, reopened, amended, duplicate and related follow-up claims.
- Healthcare insurance claims including medical, hospital, pharmacy, dental and vision claims.
- Banking and financial claims including debit or credit disputes, ATM disputes, unauthorised activity, stop-payment matters, payment disputes and fraud investigations.
- Property, casualty, travel, warranty, liability and other insurance claims.
- Straight-through, assisted, specialist-investigation and human-decision processing paths.
- Claims involving structured records, documents, correspondence, attachments, transaction history, policy or benefit data, device metadata, address information or other evidence.

## 1. Core Claim Lifecycle

Every claim shall progress through a controlled lifecycle. A claim type may skip or repeat a step only when permitted by business rules, and the reason for each transition must be retained.

| Stage | Purpose | Typical Output |
|---|---|---|
| 1. Intake and Registration | Receive and create the claim record. | Claim ID, source, party and received timestamp |
| 2. Data Collection | Gather required structured and unstructured information, including location data where relevant. | Complete claim package or missing-data list |
| 3. Validation | Confirm identity, eligibility, coverage, event validity, data and geographic consistency. | Validation result and exceptions |
| 4. Classification and Routing | Determine category, complexity, risk, priority and processing path. | Claim type, risk and queue |
| 5. Assessment / Investigation | Review evidence, rules, patterns and location anomalies. | Findings, evidence and rationale |
| 6. Remediation / Information Request | Resolve gaps or obtain evidence. | Updated package or pending status |
| 7. Decision / Adjudication | Determine outcome under governing rules. | Disposition / decision |
| 8. Financial Processing | Calculate payment, adjustment, recovery or offset. | Financial outcome |
| 9. Approval / Quality Review | Apply authority and quality controls. | Approved, returned, escalated or rejected |
| 10. Communication | Notify relevant parties. | Letter, message, explanation or request |
| 11. Closure | Complete updates and formally close. | Closure reason and audit trail |
| 12. Post-Processing | Monitor recoveries, appeals, quality and trends. | Metrics and corrective actions |

# 2. Roles and Responsibilities

| Role | Responsibility |
|---|---|
| Claim Intake / Operations | Register claims, verify basic information, manage queues and ensure mandatory fields are present. |
| Claims Processor | Review information, perform validations, gather evidence and execute authorised processing steps. |
| Location Intelligence Agent / Service | Extract and geocode addresses; correlate registered, submission, provider, laboratory, service, transaction and device locations; produce traceable geographic indicators. |
| Specialist / Investigator | Handle complex, suspicious, disputed, geographic-anomaly or domain-specific investigations. |
| Decision Maker / Adjudicator | Apply policy, benefit, product and business rules to determine the outcome. |
| Claims Manager / Approver | Review elevated authority, exceptions and manual approvals. |
| Finance / Settlement Operations | Execute approved adjustments, recoveries and payments. |
| Quality / Compliance Review | Validate adherence, evidence, authority limits and applicable obligations. |
| Customer / Member / Claimant | Provide information, documents and responses when requested. |
| Systems / Automation | Perform approved deterministic checks, retrieval, calculations, notifications and repeatable tasks. |

# 3. Claim Intake and Registration

## 3.1 Sources

Online portal, mobile application, branch or contact centre, provider submission, EDI/API, email, correspondence, system-generated referral, partner channel or manual entry. Related cases and historical claims may be linked when they concern the same event, party, account, policy, provider or transaction.

## 3.2 Minimum Registration Data

- Unique claim identifier.
- Customer, member or claimant identifier and contact information.
- Policy, plan, product, account or contract identifier where applicable.
- Claim type, category and reported reason.
- Event date/time and received date/time.
- Claimed amount or estimated exposure where applicable.
- Source channel and submitting party.
- Initial event description.
- Attached documents and evidence references.
- Address and location fields available at submission, including declared event or service location where applicable.
- Current status, owner and service-level target.

## 3.3 Intake Controls

- Check for a matching or duplicate claim.
- Validate mandatory fields before active processing.
- Record originally submitted information separately from later corrections.

- Preserve source documents, address evidence and metadata required for auditability.
- Assign the appropriate queue based on initial classification.

## 4. Data Collection and Completeness Assessment

The processor shall gather the information necessary to make a supported decision. Required data varies by domain.

| Data Category | Healthcare Example | Banking / Fraud Example | General Claim Example |
|---|---|---|---|
| Party | Member, provider | Customer, merchant, account holder | Claimant, insured, beneficiary |
| Contract / Coverage | Plan, benefit, network, authorisation | Account or product terms, dispute rights | Policy, warranty, contract |
| Event | Service, diagnosis, admission, procedure | Transaction, ATM event, payment | Loss, incident, service event |
| Financial | Billed and allowed amount, cost share | Transaction amount, adjustment, recovery | Claimed amount, excess, settlement |
| Evidence | Clinical and billing records | Transaction history, case notes, documents | Photos, receipts, reports, correspondence |
| History | Prior claims and utilisation | Prior disputes and account activity | Prior incidents and claims |
| Location Intelligence | Member address, provider, laboratory, facility and service locations | Customer, merchant, ATM, submission origin and device geolocation | Claimant, incident, submission and service-provider locations |

Geographic information may be derived from claim documents, submitted addresses, system metadata, provider records, authorised device information, transaction channels or other approved sources. Access and use shall follow applicable privacy, security and data-governance requirements.

- Complete: required information is available and the claim may proceed.
- Conditionally complete: processing may continue with a documented exception.
- Incomplete: additional information is required before decision.
- Unverifiable: information conflicts or cannot be validated; route for investigation or specialist review.

## 5. Validation and Eligibility Checks

- Identity and party validation.
- Policy, plan, product or account status and effective dates.
- Coverage, benefit, authorisation or dispute eligibility.
- Event or transaction date and applicable time limits.
- Duplicate or related-claim screening.
- Required document and evidence checks.
- Data consistency across source systems.
- Amount and currency validation.
- Existing holds, restrictions, recoveries, prior payments or offsets.
- Service-level and escalation requirements.
- Location consistency across registered party records, submission origin, service location, provider, laboratory, merchant, ATM, device and historical activity where applicable.

A failed validation or significant location discrepancy shall not automatically mean denial. The system shall distinguish a definitive rule failure, a correctable data issue, missing evidence and an exception requiring human review. Geographic anomalies must be evaluated with supporting evidence, historical activity and applicable rules.

# 6. Claim Classification and Routing

| Dimension | Examples | Routing Use |
|---|---|---|
| Claim Type | Medical, pharmacy, ATM, debit fraud, payment, property, liability | Select domain procedure |
| Complexity | Standard, moderate, complex | Determine specialist involvement |
| Risk | Low, medium, high | Increase review controls |
| Evidence State | Complete, missing, conflicting | Select remediation or investigation |
| Location Risk | Consistent, unexplained mismatch, high-risk anomaly | Continue, validate or route to investigation |
| Financial Exposure | Low, moderate, high | Apply authority thresholds |
| Urgency | Routine, urgent, time-sensitive | Prioritise queue and SLA |
| Outcome Path | Approve, deny, partially approve, recover, escalate, request information | Select next stage |

Classification shall be based on explicit rules and available evidence. Contextual or location analysis may support classification, but the outcome must remain traceable to the information and rules used.

# 7. Assessment and Investigation

## 7.1 Common Investigation Pattern

1. Establish the claim question and what must be proved before decision.
2. Review the party statement and submitted evidence.
3. Retrieve relevant authorised source records.
4. Compare facts with policy, plan, contract, product and business rules.
5. Identify inconsistencies, missing evidence, duplicates, unusual patterns and exception indicators.
6. Perform location-intelligence and geographic-consistency analysis when location data is available.
7. Perform other domain-specific checks.
8. Document findings, evidence references and rationale.
9. Determine whether additional information, specialist review or human approval is required.

### 7.1.1 Location Intelligence and Geographic Consistency Analysis

- Extract location information from claim documents, structured records, submitted addresses, transaction data, approved device metadata and authorised sources.
- Validate and geocode available addresses to establish the registered or declared geographic profile of relevant parties and entities.
- Correlate member or customer location with claim submission origin, service location, provider, laboratory, merchant, ATM, transaction and device locations as applicable.
- Compare current locations with historical claim or transaction activity and expected behavioural patterns.
- Identify significant distance, unexpected cross-region activity, improbable travel, service-location mismatch, unusual provider relationships, high-risk geography or proxy/VPN indicators where approved data supports such analysis.
- Evaluate whether anomalies have a reasonable medical, travel, operational, relocation or business explanation.
- Record the locations compared, source and timestamp, geocoding result, distance or pattern finding, rule outcome, evidence reference and risk indicator.
- Route significant unexplained anomalies to a specialist investigator or human reviewer before payment approval.

Location intelligence is a supporting fraud-risk and routing signal. It shall not, by itself, determine denial, non-payment or an adverse final disposition.

### 7.2 Healthcare Branch

- Validate member eligibility and coverage for the date of service.
- Validate provider participation or network status where relevant.
- Review procedure, diagnosis, service, authorisation and billing information.
- Apply benefit, coding, pricing, deductible, copayment and coinsurance rules.
- Resolve documentation or coding discrepancies through the defined remediation path.
- Validate consistency among member address, provider, facility, laboratory, service and submission locations. Investigate significant unexplained geographic relationships or travel patterns inconsistent with treatment history or established member activity.

### 7.3 Banking / Fraud / ATM Branch

- Validate account and transaction information.
- Review channel, entry mode, merchant or ATM information and customer statements.
- Check authorisation indicators, duplicates, account history and applicable fraud or dispute rules.
- Apply the approved investigation procedure for the relevant dispute category.
- Assess recovery rights, credit requirements and applicable business or regulatory rules.
- Correlate customer, transaction, merchant, ATM, submission-origin and device locations with historical activity. Identify unusual geography, impossible travel, location mismatch, anomalous device usage or high-risk geographic indicators and route significant findings for specialist investigation.

### 7.4 Other Claim Branches

- Property: inspect loss details, ownership, coverage, incident location, damage evidence, estimates and recoveries.
- Liability: establish incident facts, involved parties, location, responsibility indicators and evidence.
- Travel: validate itinerary, location sequence, covered event, expenses and receipts.
- Warranty: validate product, warranty period, service location, failure description and repair history.

## 8. Remediation and Additional Information

- Correct a data issue only from an approved source and retain the original value for audit.
- Request missing evidence with a clear explanation of what is required and why.
- Set the appropriate pending status and record the request and response deadline.
- Re-run relevant validation, including geographic checks, when new information is received.
- Escalate unresolved conflicts or unexplained location anomalies to the designated reviewer.

## 9. Decision / Adjudication

- Base the decision on documented facts and applicable rules.
- Reference the relevant rule, policy or benefit provision, evidence and investigation finding.
- Ensure an authorised decision maker or approved deterministic rule governs the final outcome.
- Check authority limits before final disposition.
- Record the approving role and reason for any exception.
- Do not use a location indicator as the sole basis for denial or adverse disposition.

## 10. Financial Processing and Settlement

- Calculate payable, recoverable or adjusted amounts using approved formulas and rules.
- Apply applicable deductibles, cost share, fees, interest, offsets, recoveries or chargebacks.
- Validate calculations before posting or releasing funds.
- Ensure settlement destination and payment method are authorised.
- Record inputs, outputs and adjustment reasons.
- Maintain payment hold or pending approval when investigation of a high-risk anomaly remains open.

## 11. Approval and Quality Review

- Apply monetary, risk, exception and authority thresholds.

- Perform quality review for selected claims, risk rules, exceptions and mandatory criteria.
- Verify evidence supports the decision and required updates are complete.
- Confirm high-risk geographic findings have documented human review before final approval.
- Return incomplete claims for correction and escalate material exceptions.

## 12. Customer / Member / Claimant Communication

- Acknowledge receipt where applicable.
- Request missing information with clear instructions and response expectations.
- Provide required status updates.
- Communicate the final decision, amount and understandable explanation.
- Provide applicable review, appeal, dispute or reconsideration information.
- Record channel, timestamp, template version and delivery status.
- Do not disclose sensitive fraud-detection logic, device intelligence or unnecessary location detail in routine communications.

## 13. Claim Closure

- Confirm all required stages are complete.
- Confirm final disposition and financial outcome are recorded.
- Confirm required communications are generated or sent.
- Resolve or formally disposition open tasks, holds and pending requests.
- Complete audit, quality and location-intelligence fields where applicable.
- Close with a standardised reason.
- Retain records under the applicable retention schedule.

### 9.1 Decision Types

| Decision Type | Meaning |
|---|---|
| Approved / Pay | Requirements are satisfied and payment or benefit is authorised. |
| Partially Approved | Only part of the amount or benefit is supported. |
| Denied / Not Payable | Requirements are not satisfied under governing rules. |
| Duplicate | The claim duplicates a processed or active claim. |
| Withdrawn / Cancelled | The claim is withdrawn or cancelled through an authorised process. |
| Pending Information | A final decision awaits required information. |
| Escalated | Specialist, management, compliance or higher-level review is required. |
| Reopened | A closed claim returns to active processing under an authorised reason. |

## 14. Exception, Escalation and Human Review Framework

| Trigger | Required Action |
|---|---|
| Conflicting evidence | Route to specialist or manual review. |
| Missing mandatory evidence | Request information and place claim in pending status. |
| Rule exception | Escalate to the authorised exception owner. |
| High financial exposure | Apply elevated approval authority. |

| Trigger | Required Action |
|---|---|
| Potential fraud or unusual activity | Route to the designated investigation process. |
| Geographic inconsistency or location anomaly | Validate sources and explanations; route significant unexplained anomalies for specialist fraud assessment. |
| Low confidence or ambiguous interpretation | Require human validation before final decision. |
| System or data source unavailable | Use the approved contingency process; do not fabricate missing information. |
| Customer dispute or appeal | Route to the applicable reconsideration or complaint process. |
| Regulatory or compliance-sensitive condition | Route to the designated compliance or legal procedure. |

## 15. Standard Status Model

| Status | Usage |
|---|---|
| Received | Claim received but not yet processed. |
| In Validation | Mandatory checks are being performed. |
| In Assessment | Standard assessment is under way. |
| Under Investigation | Specialised, fraud or geographic-anomaly investigation is in progress. |
| Pending Information | Additional information has been requested. |
| Pending Approval | Decision or settlement requires authorisation. |
| Approved | Decision has been approved. |
| Denied | Claim is not payable. |
| Partially Approved | Claim is payable in part. |
| Payment / Settlement in Progress | Financial processing is under way. |
| Escalated | Claim is routed to a higher-level or specialist queue. |
| Reopened | A closed claim is active again. |
| Closed | All required processing is complete. |

## 16. Evidence and Audit Requirements

- Maintain a complete chronological activity history.
- Record who or what performed each material action, including automated actions.
- Retain source documents and evidence references.
- Record rule outcomes, calculations, decision rationale and approval history.
- Maintain before-and-after values for material corrections.
- Link relevant claims, transactions, policies, accounts, providers, members and entities.
- For location analysis, retain the extracted address or location, source, timestamp, normalised or geocoded result, compared locations, finding, rule outcome and reviewer action.
- Ensure sensitive personal and location information is accessed only by authorised roles and systems.
- Maintain traceability from intake through closure.

## 17. Generalised Processing Decision Flow

| Step | Question | If Yes | If No |
|---|---|---|---|
| 1 | Is a valid claim record present? | Continue | Create or repair intake record |
| 2 | Is required data available? | Validate | Request information / pending |
| 3 | Does the claim pass deterministic validation? | Classify | Resolve, deny if definitive, or escalate |
| 4 | Are location data and relationships consistent where relevant? | Continue assessment | Validate explanation or investigate |
| 5 | Can the standard path handle the claim? | Assess / investigate | Specialist routing |
| 6 | Is evidence sufficient? | Adjudicate | Request evidence / investigate |
| 7 | Is the decision within authority? | Financial processing | Approval / escalation |
| 8 | Is settlement complete? | Communicate | Resolve settlement exception |
| 9 | Are closure conditions satisfied? | Close | Complete outstanding actions |

## 18. Domain Extension Model

Each domain extension shall define:

- Claim taxonomy: types, subtypes and disposition codes.
- Required data model: mandatory and optional fields, including applicable addresses and locations.
- Validation rules: deterministic eligibility, coverage, timing, duplicate and geographic-consistency checks.
- Investigation rules: evidence sources, fraud indicators and domain checks.
- Location-intelligence rules: address extraction, geocoding, geographic correlation, proximity validation, submission-origin analysis, historical comparison, anomaly thresholds, permissible sources and review requirements.
- Decision rules: approval, partial approval, denial and escalation conditions.
- Financial rules: calculation, adjustment, recovery and settlement logic.
- Authority matrix: monetary and exception approval levels.
- Communication templates: intake, information request, decision and closure.
- SLA model: processing targets and escalation timers.
- Audit requirements: evidence, events, privacy controls and retention.

## 19. Automation and Assisted Processing Principles

The SOP supports a hybrid operating model. Repeatable deterministic activities should remain governed by explicit rules. Interpretation of documents, narratives, relationships or conflicting evidence may use contextual and location analysis, subject to validation and human oversight.

| Activity | Preferred Pattern | Control |
|---|---|---|
| Data retrieval | System integration / deterministic | Source and timestamp |
| Mandatory validation | Rules-based | Rule result |
| Duplicate screening | Rules plus matching | Matched records and rationale |

| Activity | Preferred Pattern | Control |
|---|---|---|
| Claim classification | Rules plus contextual analysis | Classification evidence |
| Document summarisation | Contextual analysis | Source-document references |
| Address extraction / geocoding | Controlled location service | Source, normalised result and confidence |
| Geographic correlation | Rules plus location analysis | Compared entities, finding and rationale |
| Specialised investigation | Rules plus contextual analysis | Evidence and findings |
| Disposition selection | Rules / authorised decision | Applicable rule and authority |
| Financial calculation | Deterministic | Formula and inputs |
| Final approval | Human or authorised rule | Authority record |

## 20. Operational Metrics

- Claims received by type, channel and period.
- Average and percentile processing time.
- Straight-through processing rate.
- Pending-information rate and ageing.
- Escalation and specialist-review rate.
- Approval, denial, partial approval and other disposition volumes.
- Location-data availability, geographic-anomaly referral and false-positive review outcomes, subject to governance approval.
- Reopen and correction rates.
- Financial amount processed, adjusted, recovered and paid.
- Quality error rate, rework rate and SLA adherence.
- Communication turnaround and recurring exception categories.

## 21. Generalised SOP: End-to-End Procedure

10. Receive the claim through an authorised channel and create or locate the claim record.
11. Capture minimum registration information and preserve submitted evidence.
12. Check for duplicate, related or previously processed claims.
13. Collect required party, contract, event, financial, evidence and location data.
14. Run mandatory validation and eligibility checks.
15. Extract and validate addresses or geographic information when relevant and authorised.
16. Classify by type, complexity, risk, urgency, evidence state and processing path.
17. Perform standard assessment and retrieve authorised records.
18. Correlate member or customer, submission, provider, laboratory, service, transaction, merchant, ATM and device locations as applicable.
19. Evaluate geographic anomalies with historical activity, supporting evidence and reasonable explanations.
20. Conduct specialised investigation and document findings where required.
21. Request missing information and set pending status when incomplete.
22. Determine disposition using applicable rules and documented evidence.
23. Check authority and escalate thresholds, exceptions and high-risk unresolved anomalies.
24. Calculate the approved, adjusted, recovered or payable amount using controlled logic.
25. Execute settlement or record the non-payment outcome.
26. Generate and deliver the applicable communication.
27. Complete quality and audit checks.
28. Close when all criteria are satisfied.
29. Monitor for recovery, appeal, complaint, correction or reopening.

## 22. Key Design Principle

One common claim lifecycle, multiple domain-specific rule sets. The domain changes the data, rules, evidence, calculations and location relationships, not the fundamental lifecycle. New claim types should be added through a domain extension rather than a separate end-to-end process.

## Appendix A - Suggested Claim Data Model

| Group | Representative Fields |
|---|---|
| Claim | claim_id, claim_type, subtype, status, disposition, received_date, source_channel, priority |
| Party | party_id, name, contact, relationship, registered_address |
| Contract | policy/plan/product/account_id, effective_dates, coverage_status |
| Event | event_date, service_date, transaction_date, declared_location, description |
| Financial | claimed, approved, adjusted, recovery, payment, currency |
| Evidence | documents, attachments, notes, source references, evidence_status |
| Location Intelligence | extracted_address, address_source, normalised_address, geocode, submission_location, provider_location, laboratory_location, merchant_or_ATM_location, device_location, distance_or_pattern_result, anomaly_indicator, explanation, reviewer_outcome |
| Investigation | findings, indicators, checks_performed, specialist_review, rationale |
| Decision | decision, reason, rule_reference, confidence_or_review_flag |
| Workflow | owner, queue, SLA_due, escalation_status, approval_level |
| Audit | created_by, updated_by, timestamps, activity_history, approval_history |

## Appendix B - Example Domain Mapping

| Common Stage | Healthcare Claim | Fraud / ATM Claim |
|---|---|---|
| Intake | Provider or member submission | Customer dispute or transaction claim |
| Data Collection | Member, provider, clinical, billing and location evidence | Account, transaction, statement, device and location records |
| Validation | Eligibility, coverage, provider, coding, authorisation and geographic consistency | Account, transaction, timing, duplicate, dispute eligibility and geographic consistency |
| Classification | Medical, pharmacy, dental and risk path | ATM, debit fraud, debit non-fraud, stop payment and risk path |
| Investigation | Clinical, billing, benefit and location assessment | Transaction, authorisation, device and location assessment |
| Decision | Pay, adjust, deny, pend or escalate | Pay, adjust, deny, duplicate or escalate |
| Financial | Allowed amount, cost share and | Adjustment, recovery, credit and |

| Common Stage | Healthcare Claim | Fraud / ATM Claim |
|---|---|---|
| | payment | payment |
| Communication | Member or provider communication | Customer dispute or outcome communication |
| Closure | Claim finalised and records retained | Claim and system records closed |

## Appendix C - Healthcare Fraud Scenario: Geographic Anomaly Review

A pathology claim is submitted for reimbursement of diagnostic services related to treatment of lung adenocarcinoma. The claim contains clinical documentation, approved procedure and diagnosis codes, and a member address in Springfield, Illinois. The Location Intelligence Agent extracts the address from the claim document and geocodes it to establish the member's registered location. The system correlates this information with the claim submission location, provider details, laboratory location, authorised device information and historical claim activity. The assessment identifies geographically distant and cross-region relationships inconsistent with the member's established profile, together with unusual location patterns and device usage. Although the medical documentation appears authentic, the unexplained geographic anomalies generate a high-risk fraud signal. The claim is moved to Under Investigation and routed to a specialist investigator for additional validation before payment approval. The location finding supplements, but does not replace, eligibility, coverage, coding, clinical and documentation review.

## Appendix D - Location Intelligence Control Checklist

- Address or location source identified and retained.
- Use of location data is authorised and proportionate.
- Address normalisation and geocoding result recorded.
- Relevant party, submission, service, provider, laboratory, transaction, merchant, ATM or device locations compared.
- Historical activity and reasonable explanations considered.
- Risk indicator and supporting rationale documented.
- High-risk unresolved anomaly routed to human review.
- Final decision based on governing rules and total evidence, not location alone.
- Sensitive location data protected and retained under applicable controls.

End of SOP - Generalized Claims Processing Standard v1.1